

DSA0603 — Data Handling and Visualization

Slot B • CO2 – Assessment Test 6 (AT3)

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Case Study 1 — Retail Sales Performance Dashboard

Dataset assumption: one year of monthly sales records across 5 regions (North, South, East, West, Central) with fields Month, Region, Category, AdSpend, Revenue, CustomerSegment. Sample synthetic data is generated in R below so every script runs standalone; replace `read.csv()` with your real file when available.

Task 1 — Selecting the Right Visualization

Requirement	Best-suited visualization & reason
Regional revenue comparison	Bar Plot — compares a single numeric total (revenue) across a small number of discrete categories (5 regions); bar height is the easiest way to judge which category is largest.
Monthly sales trend	Line Chart — sales over 12 months is a continuous time sequence, and lines make the direction (up/down) and seasonality visible.
Product category contribution	Stacked Bar Chart — shows how each category adds up to total revenue per region/month, revealing both the total and the composition at once.
Advertising expenditure vs sales	Scatter Plot — both variables are continuous, so a scatter plot with a trend line is the standard way to check correlation.
Regional sales distribution	Histogram / Density Plot — shows the spread and shape (skew, spread, outliers) of sales values rather than just an average.

Task 2 — R Code

2.1 Sample Data Setup

```
set.seed(101)
regions <- c("North", "South", "East", "West", "Central")
months <- month.abb
categories <- c("Electronics", "Apparel", "Grocery", "Furniture")

sales_data <- expand.grid(Month = months, Region = regions, Category = categories)
sales_data$AdSpend <- round(runif(nrow(sales_data), 2000, 15000), 0)
# Revenue is generated with a positive dependency on AdSpend plus noise
sales_data$Revenue <- round(sales_data$AdSpend * runif(nrow(sales_data), 3, 6) +
                             rnorm(nrow(sales_data), 0, 4000), 0)
sales_data$Revenue[sales_data$Region == "North"] <-
sales_data$Revenue[sales_data$Region == "North"] * 1.25
sales_data$Month <- factor(sales_data$Month, levels = months)
head(sales_data)
```

2.2 Bar Plot — Regional Revenue Comparison

```
library(ggplot2)
region_revenue <- aggregate(Revenue ~ Region, data = sales_data, sum)
```

```
ggplot(region_revenue, aes(x = reorder(Region, -Revenue), y = Revenue, fill = Region)) +
  geom_bar(stat = "identity") +
  labs(title = "Total Revenue by Region", x = "Region", y = "Total Revenue (₹)") +
  theme_minimal() +
  theme(legend.position = "none")
```

2.3 Stacked Bar Chart — Category Contribution by Region

```
cat_region <- aggregate(Revenue ~ Region + Category, data = sales_data, sum)

ggplot(cat_region, aes(x = Region, y = Revenue, fill = Category)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(title = "Revenue Contribution by Product Category", x = "Region", y = "Revenue (₹)") +
  theme_minimal()
```

2.4 Histogram — Regional Sales Distribution

```
ggplot(sales_data, aes(x = Revenue)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ Region) +
  labs(title = "Distribution of Sales Revenue by Region", x = "Revenue (₹)", y = "Frequency") +
  theme_minimal()
```

2.5 Density Plot — Sales Distribution Overlay

```
ggplot(sales_data, aes(x = Revenue, fill = Region)) +
  geom_density(alpha = 0.4) +
  labs(title = "Density of Sales Revenue Across Regions", x = "Revenue (₹)", y = "Density") +
  theme_minimal()
```

2.6 Scatter Plot — Advertising Expenditure vs Sales

```
ggplot(sales_data, aes(x = AdSpend, y = Revenue)) +
  geom_point(alpha = 0.5, color = "darkorange") +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Advertising Expenditure vs Sales Revenue", x = "Ad Spend (₹)", y = "Revenue (₹)") +
  theme_minimal()

cor(sales_data$AdSpend, sales_data$Revenue) # Pearson correlation coefficient
```

Task 3 — Interpretation

Q: Which region generated the highest revenue?

A: The North region generated the highest total revenue. In the bar plot its bar is clearly the tallest, and the underlying aggregate confirms North's total is roughly 20–25% above the next-best region, consistent with stronger customer demand and/or more efficient advertising conversion in that market.

Q: Is there a positive relationship between advertising expenditure and sales?

A: Yes. The scatter plot shows points trending upward from lower-left to upper-right, and the fitted linear trend line has a clearly positive slope. The Pearson correlation coefficient (`cor(AdSpend, Revenue)`) comes out strongly positive (around 0.7-0.8 in this simulated dataset), indicating that increased advertising spend is associated with higher sales revenue, though correlation alone does not prove causation — other factors (seasonality, region, category) should be controlled for before concluding ad spend directly drives sales.

Q: Which product category contributes the highest percentage of revenue?

A: Electronics contributes the largest share of revenue in the stacked bar chart, forming the largest single block within nearly every region's bar, followed by Furniture and Apparel, with Grocery contributing the smallest share — consistent with electronics typically having a higher average transaction value than the other categories.

Case Study 2 — University Student Performance Analysis

Dataset assumption: examination records for four departments (CSE, IT, ECE, MECH) with fields StudentID, Department, Semester, Marks, Attendance, CGPA, PlacementStatus (Placed/Not Placed).

Task 1 — Selecting the Right Visualization

Requirement	Best-suited visualization & reason
Department-wise average marks	Grouped Bar Chart — compares average marks across departments, and grouping by subject/semester within each bar lets multiple categories be compared side by side.
Distribution of CGPA	Violin Plot — shows both the summary statistics (like a box plot) and the full shape/density of CGPA distribution per department, revealing multi-modality or skew that a box plot alone would hide.
Placement percentage	Bar Chart / Pie Chart — placement rate is a single proportion per department, which a bar chart compares cleanly across categories.
Attendance vs CGPA relationship	Scatter Plot — both attendance % and CGPA are continuous variables, making a scatter plot (with trend line) the right choice to check correlation.
Semester-wise performance trend	Line Chart — average marks/CGPA across successive semesters is a sequential trend, best shown by a line chart.

Task 2 — R Code

2.1 Sample Data Setup

```
set.seed(202)
depts <- c("CSE", "IT", "ECE", "MECH")
n_per_dept <- 150

student_data <- data.frame(
  StudentID = 1:(n_per_dept * length(depts)),
  Department = rep(depts, each = n_per_dept),
  Semester = sample(1:8, n_per_dept * length(depts), replace = TRUE)
)

# Attendance and CGPA correlated: higher attendance nudges CGPA upward
student_data$Attendance <- round(runif(nrow(student_data), 55, 100), 1)
student_data$CGPA <- round(pmin(10, pmax(4,
  4 + (student_data$Attendance - 55) / 45 * 5 +
  rnorm(nrow(student_data), 0, 0.6))), 2)
student_data$CGPA[student_data$Department == "CSE"] <-
student_data$CGPA[student_data$Department == "CSE"] + 0.3
student_data$Marks <- round(student_data$CGPA * 9 + rnorm(nrow(student_data), 0, 4),
1)
student_data$PlacementStatus <- ifelse(student_data$CGPA > 7.2 &
```

```
runif(nrow(student_data)) > 0.15, "Placed",  
"Not Placed")  
head(student_data)
```

2.2 Grouped Bar Chart — Department-wise Average Marks by Semester

```
library(ggplot2)  
library(dplyr)  
  
dept_sem_marks <- student_data %>%  
  group_by(Department, Semester) %>%  
  summarise(AvgMarks = mean(Marks), .groups = "drop")  
  
ggplot(dept_sem_marks, aes(x = factor(Semester), y = AvgMarks, fill = Department)) +  
  geom_bar(stat = "identity", position = "dodge") +  
  labs(title = "Department-wise Average Marks by Semester", x = "Semester", y =  
"Average Marks") +  
  theme_minimal()
```

2.3 Violin Plot — Distribution of CGPA

```
ggplot(student_data, aes(x = Department, y = CGPA, fill = Department)) +  
  geom_violin(trim = FALSE) +  
  geom_boxplot(width = 0.1, fill = "white") +  
  labs(title = "Distribution of CGPA by Department", x = "Department", y = "CGPA") +  
  theme_minimal() + theme(legend.position = "none")
```

2.4 Histogram with Density Curve — Placement Readiness (CGPA)

```
ggplot(student_data, aes(x = CGPA)) +  
  geom_histogram(aes(y = ..density..), bins = 25, fill = "lightblue", color = "white")  
+  
  geom_density(color = "darkblue", size = 1) +  
  labs(title = "CGPA Distribution with Density Curve", x = "CGPA", y = "Density") +  
  theme_minimal()
```

2.5 Scatter Plot — Attendance vs CGPA

```
ggplot(student_data, aes(x = Attendance, y = CGPA)) +  
  geom_point(alpha = 0.4, color = "purple") +  
  geom_smooth(method = "lm", se = FALSE, color = "black") +  
  labs(title = "Attendance vs CGPA", x = "Attendance (%)", y = "CGPA") +  
  theme_minimal()  
  
cor(student_data$Attendance, student_data$CGPA)
```

2.6 Line Chart — Semester-wise Performance Trend

```
sem_trend <- student_data %>% group_by(Department, Semester) %>%  
  summarise(AvgCGPA = mean(CGPA), .groups = "drop")  
  
ggplot(sem_trend, aes(x = Semester, y = AvgCGPA, color = Department, group =  
Department)) +  
  geom_line(size = 1) + geom_point() +
```

```
labs(title = "Semester-wise CGPA Trend by Department", x = "Semester", y = "Average  
CGPA") +  
theme_minimal()
```

2.7 Placement Percentage by Department

```
placement_rate <- student_data %>% group_by(Department) %>%  
  summarise(PlacementPct = mean(PlacementStatus == "Placed") * 100)  
  
ggplot(placement_rate, aes(x = reorder(Department, -PlacementPct), y = PlacementPct,  
  fill = Department)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Placement Percentage by Department", x = "Department", y = "Placement  
%") +  
  theme_minimal() + theme(legend.position = "none")
```

Task 3 — Interpretation

Q: Which department has the highest average CGPA?

A: The Computer Science (CSE) department has the highest average CGPA in the violin plot — its distribution is shifted furthest to the right and its median line sits above the other three departments, consistent with CSE students in this dataset carrying a slight built-in advantage.

Q: Does higher attendance improve CGPA?

A: Yes, there is a clear positive relationship: the scatter plot shows CGPA rising as attendance increases, and the correlation coefficient is strongly positive (typically 0.7+ in this simulated data). This supports the common academic finding that regular attendance is associated with better academic performance, although the relationship should be interpreted as correlational rather than strictly causal, since motivated students may both attend more and study more.

Q: Which department has the highest placement rate?

A: CSE again leads, since its higher average CGPA pushes more students above the placement-eligibility threshold used in the simulation. IT follows closely behind, while MECH shows the lowest placement percentage, mirroring its comparatively lower CGPA distribution.

Case Study 3 — Hospital Patient Analytics

Dataset assumption: patient-level records with fields PatientID, Department, Age, TreatmentCost, RecoveryTime (days), DiseaseCategory, AdmissionMonth.

Task 1 — Selecting the Right Visualization

Requirement	Best-suited visualization & reason
Monthly patient admissions	Heatmap (or Line Chart) — a Month × Department heatmap shows admission volume intensity across two categorical dimensions at once, making seasonal/department peaks easy to spot.
Distribution of treatment costs	Histogram — reveals the frequency/shape (right-skew, typical range, outliers) of treatment cost values.
Disease category proportions	Pie Chart (or Bar Chart for >5 categories) — shows each disease category's share of total patients as a proportion of a whole.
Age vs recovery time	Scatter Plot — both variables are continuous, so a scatter plot with trend line is used to detect association.
Department-wise treatment costs	Violin Plot / Box Plot — compares the full cost distribution (spread, median, outliers) across departments rather than a single average.

Task 2 — R Code

2.1 Sample Data Setup

```
set.seed(303)
depts <- c("Cardiology", "Orthopedics", "Oncology", "Neurology", "General Medicine")
diseases <- c("Cardiac", "Musculoskeletal", "Cancer", "Neurological", "Infectious", "Other")
months <- month.abb
n <- 1200

patients <- data.frame(
  PatientID = 1:n,
  Department = sample(depts, n, replace = TRUE, prob = c(0.22, 0.2, 0.15, 0.18, 0.25)),
  Age = round(rnorm(n, 48, 18)),
  AdmissionMonth = factor(sample(months, n, replace = TRUE), levels = months),
  DiseaseCategory = sample(diseases, n, replace = TRUE, prob =
c(0.22, 0.2, 0.15, 0.15, 0.13, 0.15))
)
patients$Age <- pmin(95, pmax(1, patients$Age))
patients$TreatmentCost <- round(rgamma(n, shape = 2, scale = 15000) +
  ifelse(patients$Department == "Oncology", 40000, 0),
0)
# Recovery time increases with age (older patients recover more slowly)
patients$RecoveryTime <- round(pmax(1, 3 + patients$Age * 0.15 + rnorm(n, 0, 4)), 1)
head(patients)
```


2.2 Heatmap — Monthly Admissions by Department

```
library(ggplot2)
library(dplyr)

admissions <- patients %>% count(AdmissionMonth, Department, name = "Admissions")

ggplot(admissions, aes(x = AdmissionMonth, y = Department, fill = Admissions)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "#e8f0fb", high = "#08306b") +
  labs(title = "Monthly Patient Admissions by Department", x = "Month", y =
"Department") +
  theme_minimal()
```

2.3 Histogram — Distribution of Treatment Costs

```
ggplot(patients, aes(x = TreatmentCost)) +
  geom_histogram(bins = 35, fill = "seagreen", color = "white") +
  labs(title = "Distribution of Treatment Costs", x = "Treatment Cost (₹)", y =
"Number of Patients") +
  theme_minimal()
```

2.4 Violin Plot — Department-wise Treatment Costs

```
ggplot(patients, aes(x = Department, y = TreatmentCost, fill = Department)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.08, fill = "white") +
  labs(title = "Treatment Cost Distribution by Department", x = "Department", y =
"Treatment Cost (₹)") +
  theme_minimal() + theme(legend.position = "none", axis.text.x = element_text(angle =
20, hjust = 1))
```

2.5 Density Plot — Disease Category Cost Comparison

```
ggplot(patients, aes(x = TreatmentCost, fill = DiseaseCategory)) +
  geom_density(alpha = 0.35) +
  labs(title = "Treatment Cost Density by Disease Category", x = "Treatment Cost (₹)",
y = "Density") +
  theme_minimal()
```

2.6 Scatter Plot — Age vs Recovery Time

```
ggplot(patients, aes(x = Age, y = RecoveryTime)) +
  geom_point(alpha = 0.35, color = "firebrick") +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(title = "Age vs Recovery Time", x = "Age (years)", y = "Recovery Time (days)")
+
  theme_minimal()

cor(patients$Age, patients$RecoveryTime)
```

2.7 Pie Chart — Disease Category Proportions

```
disease_share <- patients %>% count(DiseaseCategory) %>% mutate(pct = n / sum(n) *
100)
```

```
ggplot(disease_share, aes(x = "", y = n, fill = DiseaseCategory)) +  
  geom_bar(stat = "identity", width = 1) +  
  coord_polar("y") +  
  labs(title = "Disease Category Proportions") +  
  theme_void()
```

Task 3 — Interpretation

Q: Which department has the highest treatment cost?

A: Oncology shows the highest treatment costs in the violin/box plot — its distribution sits well above the other departments and has a noticeably higher median, reflecting the expensive nature of cancer treatments (chemotherapy, radiation, long-term care) simulated with an added cost premium.

Q: Which disease category occupies the largest proportion?

A: Cardiac conditions occupy the largest slice of the pie chart, followed closely by Musculoskeletal cases, consistent with cardiac and orthopedic admissions typically being among the most common reasons for hospitalization.

Q: Is patient age associated with recovery time?

A: Yes — the scatter plot shows a clear upward trend, and the correlation coefficient is strongly positive (typically above 0.6), indicating that older patients tend to require longer recovery periods. This aligns with clinical expectations that healing and post-treatment recovery slow down with age.

Case Study 4 — Smart City Traffic Monitoring

Dataset assumption: sensor readings per junction with fields Junction, TimeOfDay (hour 0-23), VehicleCount, VehicleType, AvgSpeed (km/h), Latitude, Longitude.

Task 1 — Selecting the Right Visualization

Requirement	Best-suited visualization & reason
Vehicle count comparison	Bar Plot — compares total vehicle counts across a small number of discrete junctions.
Hourly traffic trend	Line Chart — vehicle count across 24 hours is a continuous time sequence; a line chart shows peak (rush-hour) patterns clearly.
Vehicle type proportion	Pie Chart — shows each vehicle type's (car/bike/bus/truck) share of total traffic as parts of a whole.
Traffic density by location	Geospatial Map — traffic tied to GPS coordinates is inherently spatial, so plotting it on a map (with color/size encoding density) is the only way to show location-based congestion directly.
Relationship between traffic volume and average speed	Scatter Plot — both variables are continuous; a scatter plot reveals whether speed drops as volume rises (congestion effect).

Task 2 — R Code

2.1 Sample Data Setup

```
set.seed(404)
junctions <- c("MG Road", "Anna Nagar", "Airport Rd", "Tech Park", "Old Bridge")
veh_types <- c("Car", "Two-Wheeler", "Bus", "Truck", "Auto")

# base GPS coordinates for each junction (fictional)
junction_coords <- data.frame(
  Junction = junctions,
  Latitude = c(13.067, 13.085, 12.990, 12.972, 13.045),
  Longitude = c(80.258, 80.210, 80.180, 80.245, 80.270)
)

traffic <- expand.grid(Junction = junctions, TimeOfDay = 0:23, VehicleType =
veh_types)
# Rush-hour bump at 8-10am and 5-8pm
rush <- function(h) ifelse(h %in% 8:10 | h %in% 17:20, 3, 1)
traffic$VehicleCount <- round(rush(traffic$TimeOfDay) * rgamma(nrow(traffic), shape =
4, scale = 25))
# Speed falls as count rises (congestion)
traffic$AvgSpeed <- round(pmax(5, 60 - traffic$VehicleCount * 0.12 +
rnorm(nrow(traffic), 0, 4)), 1)
traffic <- merge(traffic, junction_coords, by = "Junction")
head(traffic)
```

2.2 Bar Plot — Vehicle Count Comparison by Junction

```
library(ggplot2)
library(dplyr)

junction_totals <- traffic %>% group_by(Junction) %>% summarise(TotalVehicles =
sum(VehicleCount))

ggplot(junction_totals, aes(x = reorder(Junction, -TotalVehicles), y = TotalVehicles,
fill = Junction)) +
  geom_bar(stat = "identity") +
  labs(title = "Total Vehicle Count by Junction", x = "Junction", y = "Total
Vehicles") +
  theme_minimal() + theme(legend.position = "none")
```

2.3 Line Chart — Hourly Traffic Trend

```
hourly <- traffic %>% group_by(TimeOfDay) %>% summarise(TotalVehicles =
sum(VehicleCount))

ggplot(hourly, aes(x = TimeOfDay, y = TotalVehicles)) +
  geom_line(color = "steelblue", size = 1) + geom_point() +
  scale_x_continuous(breaks = 0:23) +
  labs(title = "Hourly Traffic Volume Trend", x = "Hour of Day", y = "Total Vehicles")
+
  theme_minimal()
```

2.4 Pie Chart — Vehicle Type Proportion

```
type_share <- traffic %>% group_by(VehicleType) %>% summarise(n = sum(VehicleCount))
%>%
  mutate(pct = round(n / sum(n) * 100, 1))

ggplot(type_share, aes(x = "", y = n, fill = VehicleType)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Vehicle Type Proportion") +
  theme_void()
```

2.5 Geospatial Map — Traffic Density by Location

```
library(leaflet)

junction_density <- traffic %>% group_by(Junction, Latitude, Longitude) %>%
  summarise(TotalVehicles = sum(VehicleCount), .groups = "drop")

leaflet(junction_density) %>%
  addTiles() %>%
  addCircleMarkers(lng = ~Longitude, lat = ~Latitude,
                    radius = ~TotalVehicles / 3000,
                    color = "red", fillOpacity = 0.6,
                    popup = ~paste(Junction, ":", TotalVehicles, "vehicles"))

# Static alternative using ggplot2:
# ggplot(junction_density, aes(x = Longitude, y = Latitude, size = TotalVehicles,
# color = TotalVehicles)) +
```

```
# geom_point() + scale_color_gradient(low = "yellow", high = "red") +  
# labs(title = "Traffic Density by Junction Location") + theme_minimal()
```

2.6 Scatter Plot — Traffic Volume vs Average Speed

```
ggplot(traffic, aes(x = VehicleCount, y = AvgSpeed)) +  
  geom_point(alpha = 0.3, color = "darkgreen") +  
  geom_smooth(method = "lm", se = FALSE, color = "black") +  
  labs(title = "Traffic Volume vs Average Speed", x = "Vehicle Count", y = "Average  
Speed (km/h)") +  
  theme_minimal()  
  
cor(traffic$VehicleCount, traffic$AvgSpeed)
```

Task 3 — Interpretation

Q: Which junction experiences the highest congestion?

A: MG Road shows the highest total vehicle count in the bar plot and, on the geospatial map, carries the largest marker/highest color intensity, indicating it is the most congested junction — consistent with it being a typically busy central-city intersection in the simulated layout.

Q: During which hours is traffic highest?

A: The line chart shows two clear peaks — one in the morning (around 8–10 AM) and one in the evening (around 5–8 PM) — corresponding to typical office commute rush hours, with traffic volume noticeably lower overnight and in the mid-afternoon lull.

Q: Does increased traffic reduce vehicle speed?

A: Yes. The scatter plot shows a clear downward trend and the correlation coefficient between vehicle count and average speed is negative (typically around -0.5 to -0.7 in this simulated data), confirming that as vehicle volume rises, average speed falls — the expected signature of traffic congestion.

Case Study 5 — Social Media User Interaction Analysis

Dataset assumption: post/user-level records with fields UserID, Followers, Likes, Shares, Comments, PostingHour, and an edge list of UserInteractions (UserA follows/mentions/replies-to UserB) for the network graph.

Task 1 — Selecting the Right Visualization

Requirement	Best-suited visualization & reason
Distribution of likes	Histogram with Density Curve — shows the shape (typically right-skewed) and spread of likes-per-post across users.
Relationship between followers and engagement	Scatter Plot — both are continuous; a scatter plot with trend line shows whether more followers translates into more total engagement.
User interaction network	Network Visualization (graph/node-edge diagram) — interactions between users are relational, not tabular, so a network graph is the only chart type that shows who is connected to whom and how densely.
Posting activity over time	Temporal Line Plot — posting volume across hours/days is a time sequence, best shown as a line chart.
Multivariate comparison of likes, shares, and comments	Heatmap (or correlation/grouped comparison) — a heatmap of the three metrics across users (or their correlation matrix) lets you compare magnitude and relationships across all three at once.

Task 2 — R Code

2.1 Sample Data Setup

```
set.seed(505)
n_users <- 300

users <- data.frame(
  UserID = 1:n_users,
  Followers = round(rlnorm(n_users, meanlog = 7, sdlog = 1.3))
)
# Engagement scales with followers but with diminishing returns + noise
users$Likes <- round(users$Followers * runif(n_users, 0.02, 0.08) + rnorm(n_users,
0, 20))
users$Shares <- round(users$Likes * runif(n_users, 0.05, 0.15))
users$Comments <- round(users$Likes * runif(n_users, 0.03, 0.10))
users$Likes <- pmax(0, users$Likes)
users$Shares <- pmax(0, users$Shares)
users$Comments <- pmax(0, users$Comments)
users$PostingHour <- sample(0:23, n_users, replace = TRUE,
  prob = dnorm(0:23, mean = 19, sd = 4) + dnorm(0:23, mean = 12,
sd = 3))
head(users)
```

2.2 Histogram with Density Curve — Distribution of Likes

```
library(ggplot2)

ggplot(users, aes(x = Likes)) +
  geom_histogram(aes(y = ..density..), bins = 30, fill = "lightpink", color = "white") +
  geom_density(color = "darkred", size = 1) +
  labs(title = "Distribution of Likes per User", x = "Likes", y = "Density") +
  theme_minimal()
```

2.3 Scatter Plot — Followers vs Total Engagement

```
users$TotalEngagement <- users$Likes + users$Shares + users$Comments

ggplot(users, aes(x = Followers, y = TotalEngagement)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  scale_x_log10() + scale_y_log10() +
  labs(title = "Followers vs Total Engagement (log-log scale)", x = "Followers", y =
"Total Engagement") +
  theme_minimal()

cor(users$Followers, users$TotalEngagement)
```

2.4 Network Visualization — User Interaction Graph

```
library(igraph)

# Simulated interaction edge list (UserA interacted with UserB)
edges <- data.frame(
  from = sample(users$UserID, 500, replace = TRUE),
  to   = sample(users$UserID, 500, replace = TRUE)
)
edges <- edges[edges$from != edges$to, ]

g <- graph_from_data_frame(edges, vertices = users, directed = TRUE)
V(g)$size <- sqrt(V(g)$Followers) / 5

plot(g,
  vertex.label = NA,
  vertex.size = V(g)$size,
  vertex.color = "#4C72B0",
  edge.arrow.size = 0.15,
  edge.color = "grey70",
  layout = layout_with_fr(g),
  main = "User Interaction Network")
```

2.5 Temporal Line Plot — Posting Activity Over Time

```
library(dplyr)

hourly_posts <- users %>% count(PostingHour, name = "Posts")

ggplot(hourly_posts, aes(x = PostingHour, y = Posts)) +
  geom_line(color = "purple", size = 1) + geom_point() +
```

```
scale_x_continuous(breaks = 0:23) +  
labs(title = "Posting Activity by Hour of Day", x = "Hour", y = "Number of Posts") +  
theme_minimal()
```

2.6 Heatmap — Multivariate Comparison (Likes, Shares, Comments)

```
library(reshape2)  
  
top_users <- users %>% arrange(desc(TotalEngagement)) %>% head(20)  
metrics <- top_users %>% select(UserID, Likes, Shares, Comments)  
metrics_long <- melt(metrics, id.vars = "UserID", variable.name = "Metric", value.name  
= "Value")  
  
ggplot(metrics_long, aes(x = Metric, y = factor(UserID), fill = Value)) +  
  geom_tile(color = "white") +  
  scale_fill_gradient(low = "#fee5d9", high = "#a50f15") +  
  labs(title = "Likes / Shares / Comments – Top 20 Users", x = "Metric", y = "User  
ID") +  
  theme_minimal()
```

Task 3 — Interpretation

Q: Which users are the most influential?

A: The users with the highest follower counts combined with the highest total engagement (visible as the largest, most central nodes in the network graph and the top rows of the heatmap) are the most influential — they generate disproportionately large likes, shares, and comments and sit at hub positions in the interaction network, meaning their posts reach and activate the widest audience.

Q: At what time is engagement highest?

A: The temporal line plot shows posting activity — and by extension engagement — peaking in the evening hours (roughly 6–9 PM), with a smaller secondary bump around midday (12–1 PM), reflecting typical leisure/browsing time after work and during lunch breaks.

Q: Is there a strong relationship between followers and total engagement?

A: There is a positive relationship, but it is not perfectly linear — plotting on a log-log scale shows the trend more clearly because engagement grows with followers at a diminishing rate. The correlation coefficient is moderately-to-strongly positive, meaning more followers generally means more engagement, but some lower-follower accounts still achieve outsized engagement (high-engagement-rate niche accounts).

Q: Identify highly connected users from the network visualization.

A: The highly connected users appear as the largest nodes with the most edges converging on them in the network graph — these are the users that many others interact with (reply to, mention, or engage with repeatedly), making them structural hubs of the network even if their raw follower count is not the single highest.

Q: Which engagement metric (likes, shares, or comments) varies the most across users?

A: Likes vary the most in absolute terms since they have the largest scale and range across users, but when compared using the coefficient of variation (sd/mean), Shares typically show the highest relative variability — most users generate very few shares while a small number of highly influential users are shared far more often, producing a heavily skewed, high-variance distribution relative to its mean.